

1. (80 points) Using Tables 3-1 and 3-2, or by direct calculation, determine the following:

- a. The Laplace transform of $(t-2)u(t)$
- b. The inverse Laplace transform of $\frac{e^{-s}}{s^2}$
- c. The Laplace transform of $(t-2)e^{-t}u(t)$
- d. The Laplace transform of $(t-1)e^{-(t-1)}u(t-1)$
- e. The inverse Laplace transform of $\frac{1}{s^2+16}$
- f. The Laplace transform of $e^{-2t} \cos(3t+135^\circ)u(t)$
- g. The Laplace transform of $\frac{d}{dt}\{t^2 e^{-t}u(t)\}$
- h. The Laplace transform of $\frac{d^3 x(t)}{dt^3}$, for a causal signal $x(t)$
- i. The Laplace transform of $r(t-2)$, where $r(t)$ represents the unit ramp signal
- j. The Laplace transform of $(t-1)^2 e^{-2(t-1)} \cos(5(t-1))u(t-1)$

2. (10 points) Calculate the initial and final values for the system $H(s) = \frac{10s+3}{s^2+3s+4}$

3. (10 points) Calculate the initial and final values for the system $H(s) = \frac{s+4}{s^2}$

a. The Laplace transform of $(t-2)u(t)$

$$t u(t) - 2 u(t)$$

$$\frac{1}{s^2} - \frac{2}{s}$$

b. The inverse Laplace transform of $\frac{e^{-s}}{s^2}$

$$(t-1)u(t-1)$$

c. The Laplace transform of $(t-2)e^{-t}u(t)$

$$t e^{-t} u(t) - 2 e^{-t} u(t)$$

$$\frac{1}{(s+1)^2} - \frac{2}{s+1}$$

d. The Laplace transform of $(t-1)e^{-(t-1)}u(t-1)$

$$\frac{e^{-s}}{s+1}$$

e. The inverse Laplace transform of $\frac{1}{s^2+16}$

$$\frac{1}{4} \sin(4t) u(t)$$

f. The Laplace transform of $e^{-2t} \cos(3t+135^\circ)u(t)$

$$\frac{(s+2) \cos(135^\circ) + 3 \sin(135^\circ)}{(s+2)^2 + 3^2}$$

g. The Laplace transform of $\frac{d}{dt}\{t^2 e^{-t}u(t)\}$

$$\frac{2s}{(s+1)^3}$$

h. The Laplace transform of $\frac{d^3 x(t)}{dt^3}$, for a causal signal $x(t)$

$$s^3 X(s) - s^2 x(0^-) - s x'(0^-) - x''(0^-)$$

i. The Laplace transform of $r(t-2)$, where $r(t)$ represents the unit ramp signal

$$(t-2)u(t-2) \Rightarrow \frac{e^{-2s}}{s^2}$$

j. The Laplace transform of $(t-1)^2 e^{-2(t-1)} \cos(5(t-1))u(t-1)$

$$e^{-s} \frac{d^2}{ds^2} \left[\frac{(s+2)}{(s+2)^2 + 5^2} \right]$$

2. (10 points) Calculate the initial and final values for the system $H(s) = \frac{10s+3}{s^2+3s+4}$

Initial value: $\lim_{s \rightarrow \infty} s \left(\frac{10s+3}{s^2+3s+4} \right) = \lim_{s \rightarrow \infty} \frac{10s^2+3s}{s^2+3s+4} = 10$

Final value $\lim_{s \rightarrow 0} s \left(\frac{10s+3}{s^2+3s+4} \right) = \lim_{s \rightarrow 0} \frac{10s^2+3s}{s^2+3s+4} = 0$

3. (10 points) Calculate the initial and final values for the system $H(s) = \frac{s+4}{s^2}$

Initial value $\lim_{s \rightarrow \infty} s \left(\frac{s+4}{s^2} \right) = \lim_{s \rightarrow \infty} \frac{s^2+4s}{s^2} = 1$

Final value $\lim_{s \rightarrow 0} \frac{s^2+4s}{s^2} = 0$